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Advances in biocatalytic textiles enabled by immobilized enzyme for water purification and CO_2 capture applications

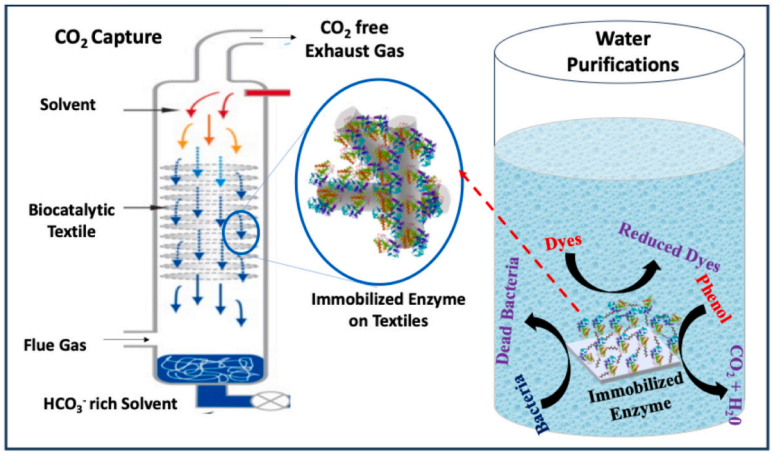
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HIGHLIGHTS

- Summarizes advances in enzyme immobilized textile for water and air purification.
- Highlights enzyme immobilization strategies shaping biocatalytic textile performance.
- Reviews immobilized enzymes in removal of pesticides, phenolics, and microbial contaminants.
- Discusses fundamentals of biocatalytic textile in CO_2 capture.
- Identifies challenges and future prospects for scalable, green remediation processes.

GRAPHICAL ABSTRACT



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ABSTRACT

Biocatalytic textiles have emerged as a new class of functional materials that merge the catalytic precision of enzymes with the structural and mechanical advantages of textiles. Their combination of high surface area, flexibility, low pressure drop, and durability enables efficient mass transfer and stable catalytic performance, positioning them as promising candidates for sustainable environmental purification technologies. This review provides an overview of the progress, opportunities, and challenges associated with biocatalytic textiles for water purification and carbon dioxide (CO_2) capture application. Starting with fundamental aspects of enzymes and enzyme immobilization strategies that underpin the design and performance of biocatalytic textiles, this report summarizes the applications of biocatalytic textiles in the removal of dyes, pharmaceuticals, pesticides, phenolic compounds and bacteria from contaminated water, demonstrating their potential for addressing key issues in wastewater treatment. Additionally, the emerging use of biocatalytic textiles for CO_2 capture is explored as a pathway toward carbon mitigation and efficient carbon management strategies. The review concludes with limitations and future research directions aimed at robust, durable, and industrially viable biocatalytic textile systems for catalytic water and air purification.

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1. Introduction

Biocatalytic textiles are advanced materials that integrate enzymes or other biocatalysts into textile structures, enabling the textiles to perform specific chemical reactions (Morshed et al., 2021a; Kiehl et al., 2015). This innovation leverages the low pressure drop, high surface area, and flexibility of textile fibers to immobilize enzymes, creating multifunctional materials for applications such as environmental remediation, green chemistry, and sustainable manufacturing (Birnbaum, 1996; Homaei, 2013; Yuan, 2021).

Biocatalytic textiles are obtained by immobilizing enzymes onto or within textile fibers, allowing the textile to catalyze chemical reactions as fluids pass through or over the material. The immobilization process transforms soluble enzymes into an insoluble, reusable form by attaching them to the textile matrix. These textile matirices can be made from both natural or synthetic fibers (Basso and Serban, 2020; Morshed, 2021a; Khan, 2021). Immobilization improves enzyme stability, and allowing repeated use. Immobilized enzymes can be easily separated from the reaction mixture, simplifying product recovery and reducing contamination (Zhong et al., 2021). They often show enhanced resistance to changes in pH and temperature and can be used in continuous processes, increasing operational efficiency (Cui et al., 2016, 2018a). The support material, such as textiles, also promotes effective enzyme–substrate contact, which may improve reaction rates and selectivity. However, enzyme immobilization has some limitations. The immobilization process may reduce enzymatic activity due to structural changes or limited substrate accessibility (Maghraby et al., 2023). Mass transfer limitation and higher initial costs of support materials can also be drawbacks. Despite these challenges, immobilization effectively overcomes major problems of free enzymes, such as poor stability, lack of reusability, and difficulty in separation, making it suitable for industrial applications (Zhong et al., 2025; Cui et al., 2018b; Wen et al., 2020).

A typical biocatalytic textile consists of a woven, knitted, nonwoven, or nanofibrous textile support coated or impregnated with a thin functional layer containing the immobilized enzyme. For example, a cellulosic yarn coated with chitosan and catalase can decompose hydrogen peroxide efficiently, with the textile's structure supporting controlled liquid transport and maximizing enzyme-substrate interactions (Yuan et al., 2021) The textile's porosity and wicking properties help direct the flow of reactants, making the system suitable for continuous flow reactors or filtration applications. These textiles are robust, washable, and can be tailored for specific reactions or environmental conditions. Biocatalytic textiles can be produced in various forms, including woven, knitted, nonwoven, and nanofibrous structures, each offering unique benefits for enzyme immobilization and catalytic performance (An, 2014; Arslan, 2011; Ibrahim, 2007; Morshed, 2021a). Applications include wastewater treatment, dye degradation, green chemical synthesis, and textile recycling, where they enable selective breakdown of complex pollutants (Ameri, 2021; An, 2014; Bayramoglu, 2012a; Morshed, 2021a, 2021b). The performance of biocatalytic textiles is influenced by robustness of enzyme immobilization. Enzyme immobilization is a complex interplay of factors including the choice of textile support, crosslinker chemistry, enzyme attachment strategy, enzyme orientation, enzyme loading, and mass transfer limitations, all of which must be carefully optimized to achieve efficient, stable, and cost-effective catalytic systems which has been extensively discussed in our earlier report (Morshed et al., 2019, 2021a, 2021b, 2021c).

Biocatalytic textiles have found wide range of applications, particularly in environmental and industrial sectors, due to their ability to immobilize enzymes and catalyze specific reactions efficiently and sustainably (Morshed et al., 2021a; Yuan et al., 2023). Biocatalytic textiles has been shown as highly effective material in degrading and decolorizing textile dyes and other organic pollutants in industrial effluents, achieving removal rates about 80–90% for various dyes and significant reductions in chemical oxygen demand (COD), biological oxygen demand (BOD), and toxicity (Cristóvão et al., 2011; Kahoush, 2019). Literature shows immobilized enzymes on textiles (referreed as biocatalytic textiles) can break down hazardous pesticides, such as organophosphates, in contaminated water, with high efficiency and reusability in both batch and continuous flow systems (Gao et al., 2014). These biocatalytic textiles can be tailored to remove pharmaceuticals and phenolic endocrine-disrupting chemicals from water, contributing to safer water supplies (Xu et al., 2013, 2017; Wang et al., 2014). More recently, the scope of biocatalytic textiles has expanded beyond liquid-phase treatment to include gas–liquid and gas-phase applications, particularly CO_2 capture. By immobilizing enzymes such as carbonic anhydrase onto textile fibers. These materials can efficiently catalyze the reversible hydration of CO_2 to bicarbonate, enabling enhanced CO_2 removal from air or industrial gas streams (Shen et al., 2022a). Notably, both water purification and CO_2 capture rely on the same underlying principles of enzyme immobilization, mass transfer control, and textile-enabled reactor design, underscoring a shared technological foundation despite differences in application environments.

This emerging field of biocatalytic textiles has evolved rapidly in recent years, accompanied by a significant rise in biotechnology exploring novel enzyme immobilization techniques, textile supports, and environmental applications (Behary, 2025; Khan, 2021). Despite the rapid advancement of biocatalytic textiles and the growing body of experimental studies reported in recent years, comprehensive and integrative reviews that systematically compile these developments and critically evaluate their implications remain limited. With this progress and the growing number of experimental studies, a number of comprehensive and integrative reviews has been published including our last review on enzyme immobilization on textiles five years ago (Morshed et al., 2021a), however they are predominantly focused on general enzyme immobilization principle, methods and influencin factors (Maghraby et al., 2023; Guisan et al., 2022; Feng et al., 2022). Despite being a foundational and highly relevant domain, the application of enzyme-immobilized textiles for water purification and CO_2 capture has not yet been comprehensively reviewed to the best of our knowledge. In this context, the present review aims to bridge this knowledge gap by providing a textile-centric and application-oriented synthesis of recent advances in biocatalytic textiles for water purification and CO_2 capture, two application domains that, despite differing operating media, share common design principles rooted in enzyme immobilization, textile architecture, and flow-through reactor concepts. By critically assessing immobilization strategies, textile structures, catalytic performance metrics, and system integration approaches, this review seeks to offer not only a consolidated overview of the field but also actionable insights into current limitations, research needs, and future opportunities for translating biocatalytic textiles into practical, real-world purification technologies.

2. Fundamentals of enzyme and enzyme immobilization

2.1. Enzymes

Enzymes are composed of chains of amino acids linked together by amide (peptide) bonds, similar to other proteins. Each polypeptide chain has a free amino group at one end (the N-terminus) and a free carboxyl group at the other (the C-terminus) (Copeland, 2023). The structural organization of enzymes can be described at four hierarchical levels: primary, secondary, tertiary, and quaternary, as illustrated in Fig. 1.

The primary structure is defined by the linear sequence of amino acids in the polypeptide chain, which dictates how the chain folds (Morshed and Mohd Yusuf, 2025; Poulos, 2014). The secondary structure arises from local interactions along the backbone, forming common motifs such as α-helices and β-sheets connected by loop regions. These elements further fold into a tertiary structure, producing a compact, three-dimensional shape stabilized by side-chain interactions, including hydrophobic packing, hydrogen bonding, and disulfide bridges (Maghraby et al., 2023; Cornish-Bowden, 2013; Nierstrasz et al., 2010).

This precise three-dimensional arrangement creates the enzyme's active site and is critical for its substrate specificity and catalytic function. Enzyme structure can extend beyond the tertiary level when multiple polypeptide chains come together, forming what is known as the quaternary structure. Quaternary structure arises when two or more folded subunits (protomers) associate through non-covalent interactions such as hydrogen bonds, hydrophobic interactions, and ionic forces, resulting in a functional holoenzyme. This structural level is not universal, as only about 30% of known enzymes display quaternary organization (Friedrich, 2014; Phillips, 1966). The quaternary arrangement defines both the stoichiometry and the three-dimensional arrangement of the subunits, which can be identical (homomeric) or different (heteromeric). Based on the reaction catalyzes, all enzymes are divided into six classes before class 7: Translocases was introduced in 2018 (Florkin et al., 1965; Tipton, 2018). The classes of enzymes are categorized in Table 1 along with function and reaction they catalyze according to various literatures (Sheikh et al., 2019; Nielsen et al., 2009; Belorkar et al., 2021).

Enzymes facilitate chemical reactions within living organisms under suitable physiological conditions. They operate through highly specific molecular interactions, binding selectively to substrate molecules and reducing the activation energy required for their transformation into products (Friedrich, 2014). Due to their remarkable catalytic efficiency, substrate specificity, and environmentally friendly nature, enzymes are increasingly considered for diverse industrial applications. The catalytic performance of enzymes depends on several interrelated factors, including temperature, pH, the ratio of enzyme to substrate, and the presence of regulatory compounds that can either enhance (activators) or inhibit (inhibitors/retarders) enzymatic activity (Engel, 2013). Enzymes achieve maximum catalytic activity within certain temperature ranges, as thermal energy affects molecular motion and, consequently, enzyme structure and function. While moderate increases in temperature can accelerate reaction rates by promoting more frequent molecular collisions, excessively high temperatures can destabilize the enzyme's tertiary structure by breaking key hydrogen bonds and ionic interactions. This structural disruption alters the geometry of the active site, ultimately causing irreversible denaturation and loss of enzymatic activity. Similarly, enzyme activity is strongly influenced by pH, with each enzyme exhibiting a specific optimal pH that ensures the correct charge states of residues critical for catalysis. Most enzymes display a bell-shaped activity profile with respect to pH, where activity declines sharply when conditions move away from the optimum due to excessive protonation at low pH or deprotonation at high pH (Copeland, 2023; Engel, 2013; Vajanapanich et al., 2025).

Table 1
Overview of enzyme class, name, function, and reaction catalyzes.

Enzyme class and Name	Function	Reaction catalyzes
EC 1: Oxidoreductases	Electron transfer reactions between molecular species, mediating biological oxidation-reduction processes.	$A_{red} + B_{ox} \rightleftharpoons A_{ox} + B_{red}$
EC 2: Transferases	Transfer of specific functional groups (e.g., methyl, phosphate, or glycosyl groups) from one substrate molecule (donor) to another (acceptor).	A – B + C → A + B – C
EC 3: Hydrolases	Bond cleavage through hydrolytic reactions, involving the nucleophilic addition of water molecules.	$A – B + H_2O \rightarrow A – H + B–OH$
EC 4: Lyases	Cleavage or formation of chemical bonds through elimination or addition reactions, without requiring hydrolysis or oxidation-reduction.	A –B ⇌ A + B (reverse reaction: synthase)
EC 5: Isomerases	Interconversion of isomers or rearrangement of atoms within a molecule.	A – B – C ⇌ A – C – B
EC 6: Ligases	Joining of two substrate molecules to form a single product, typically coupled with the hydrolysis of high-energy phosphate bonds.	$A + B + ATP \rightarrow A – B + ADP + P_i$
EC 7: Translocases	Directional transport of ions or molecules across biological membranes.	AX + B || = A + X + ||B

The relationship between substrate concentration and enzyme-catalyzed reaction rate follows key biochemical principles essential for industrial processes. At low substrate levels, reaction velocity increases proportionally with substrate concentration, as more molecules interact with available active sites to form enzyme-substrate complexes. This linear region reflects first-order kinetics, where the reaction rate is directly dependent on substrate availability (Cornish-Bowden, 2013). As substrate concentration continues to rise, the enzyme becomes saturated, all active sites are occupied, and the reaction rate reaches a maximum plateau reaching to this zero-order kinetic phase where

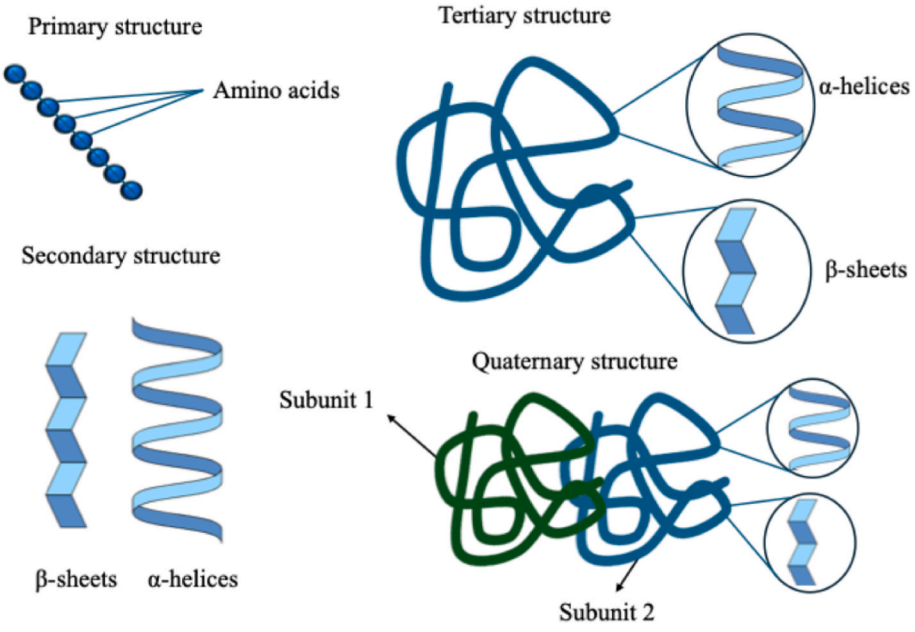


Fig. 1. Schematic illustration of primary, secondary, tertiary, and quaternary structure of proteins.

further increases in substrate do not enhance the rate because enzyme availability is limiting. This behavior is quantitatively described by the Michaelis-Menten equation, which defines the substrate concentration at half-maximal velocity as a key indicator of enzyme-substrate affinity (Engel, 2013; Lopina, 2017).

2.2. Enzyme immobilization

Enzyme immobilization refers to the physical attachment or confinement of enzymes onto insoluble support materials (Mulinari et al., 2020). The concept of immobilizing enzymes was first explored in the early 20th century, with major technological developments occurring during the 1950s and 1960s. The first industrial application of immobilized enzymes was reported in 1969 by Chibata and co-workers (Birnbaum, 1996) at Tanabe Seiyaku Company in Japan, who successfully employed immobilized aminoacylase for the stereoselective resolution of synthetic racemic D,L-amino acids. This milestone marked the introduction of immobilized enzyme technology on a large industrial scale (Mulinari et al., 2020; Tosa et al., 1966). Since then, a significant amount of research has been dedicated to the principles of enzyme immobilization (Maghraby et al., 2023; Almeida et al., 2022). There were established difference in free and immobilized enzymes as highlighted in Fig. 2. This approach has demonstrated its efficacy in various applications, including detergent, waste treatment, food industry processes as well as clarification and debittering, production of novel compounds in the medical, pharmaceutical, and chemical industries, biofuel and biodiesel production, and the pretreatment of lignocellulosic materials like sugarcane bagasse (Almeida et al., 2022). Various support materials, ranging from organic and inorganic substances to nano- and microporous structures including textiles, have been utilized for enzyme immobilization via different techniques (Maghraby et al., 2023).

The immobilization process can significantly influence the kinetic parameters of enzymatic catalysts. For example, the Michaelis constant (K_M) is indicative of the enzyme-substrate affinity, with a lower K_M value suggesting higher affinity. The reaction rate (V_{max}) is associated with the enzyme's maximum catalytic activity (Cornish-Bowden, 2013; Engel, 2013). A summary of the K_M and V_{max} values for both free and immobilized enzymes, as reported in comparative studies, is presented in Table 2. In many cases, enzyme immobilization does not significantly alter the product yield and enables the efficient recovery of biocatalysts. These biocatalysts can often be reused for as many as 15 cycles while retaining over 50% of their initial activity (Morshed et al., 2021a; Salah et al., 2024). However, enzyme immobilization tends to reduce both the maximum reaction velocity (V_{max}) and enzyme–substrate affinity, leading to an increase in the Michaelis constant (K_M). The decreased affinity between the enzyme and substrate can be attributed to two main factors: (i) diffusional limitations of the substrate, or alterations in enzyme conformation that lower substrate binding affinity, and (ii) the formation of the Nernst diffusion layer around the immobilized enzyme.

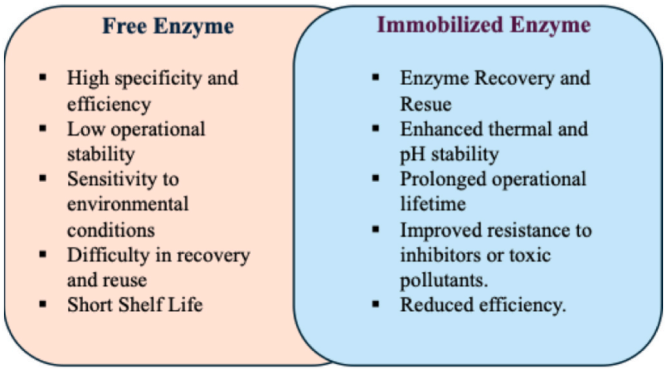


Fig. 2. Comparative representation of free and immobilized enzyme.

This layer, a thin solvent film surrounding the insoluble enzyme, creates a gradient in substrate concentration. As a result, enzyme saturation occurs at a higher substrate concentration for immobilized enzymes compared to their free, soluble counterparts. Such effects can contribute to reduced enzyme–substrate affinity and a slower reaction rate. The negative effects of immobilization on V_{max} and K_M can be overcome by optimizing the choice of support matrix, immobilization methods and overall immobilization process (Cornish-Bowden, 2013; Engel, 2013).

Enzyme immobilization methods can be broadly categorized based on the type of interaction between the enzyme and the support material, including both physical and chemical methods (as illustrated in Fig. 3) (Maghraby et al., 2023; Homaei et al., 2013). Physical methods rely on non-covalent forces such as van der Waals interactions, hydrophobic interactions, or ionic bonds that allow the enzyme to adhere to the carrier surface. This category also includes entrapment of enzymes within porous polymer networks or hydrogels, as well as microencapsulation within semi-permeable membranes. These approaches generally maintain the enzyme's native structure while providing moderate attachment strength. On the other hand, chemical immobilization involves creating covalent bonds between reactive groups on the enzyme (such as lysine ε-amino, cysteine thiol, or carboxyl groups) and chemically activated supports, resulting in stronger and more stable attachment (Maghraby et al., 2023; Robescu et al., 2025).

Enzyme immobilization can also be divided into two main types based on the strength and nature of enzyme-support interactions: reversible and irreversible methods (as illustrated in Fig. 3) (Robescu et al., 2025). Reversible immobilization relies on non-covalent forces, such as physical adsorption, ionic interactions, or affinity binding, allowing the enzyme to be detached from the support under specific conditions without losing activity. This feature enables the reuse of supports and recovery of enzymes when needed. In contrast, irreversible immobilization creates a permanent attachment between the enzyme and the support, either through covalent bonds or entrapment within polymeric matrices, resulting in highly stable biocatalytic systems that cannot be separated without affecting enzyme function. The choice of method depends on the intended application: reversible techniques offer flexibility for systems requiring periodic enzyme replacement, whereas irreversible approaches provide enhanced stability for continuous industrial operations. Overall, the most widely used immobilization methods include (a) physical adsorption, (b) covalent binding, (c) crosslinking, (d) entrapment, and (e) encapsulation (Guisan et al., 2020; Sirisha et al., 2016).

Physical adsorption for enzyme immobilization depends on non-covalent interactions between the enzyme and the support surface, such as electrostatic attractions, hydrophobic effects, van der Waals forces, and hydrogen bonds. This process generally follows the Langmuir isotherm, where the amount of enzyme adsorbed is influenced by enzyme concentration and binding affinity. Physical adsorption allows reversible attachment, preserving the enzyme's native structure and catalytic activity (Hudson et al., 2005). In contrast, covalent binding is a more permanent immobilization method, forming strong chemical bonds between enzyme molecules and carrier surfaces. Typically, reactive functional groups on both the enzyme (e.g., amino, carboxyl, or hydroxyl groups) and the support are activated and then linked using bifunctional reagents such as glutaraldehyde, carbodiimides, or epoxides. This approach not only stabilizes the enzyme by limiting conformational flexibility but can also enhance stereoselectivity by properly orienting the active site (Sirisha et al., 2016). A related strategy, cross-linking immobilization, connects enzyme molecules through similar chemical reactions, either in the presence or absence of a support material. Homobifunctional cross-linkers, such as dimethyl suberimidate or bis-diazobenzidine, form intermolecular bonds between lysine or cysteine residues of neighboring enzymes, producing stable enzyme aggregates. Notably, carrier-free cross-linked enzyme aggregates (CLEAs) achieve high volumetric activity by eliminating non-catalytic support material (Maghraby et al., 2023; Sirisha et al.,

Table 2
Comparative analysis of kinetic parameters of free and immobilized enzymes.

Enzyme	V_{max} (f)	V_{max} (i)	K_m (f)	K_m (i)	Reuse Cycles (i)	Ref.
Glucose Oxidase	364 μmol/min.ml	261 μmol/min.ml	2.7 mM	2.9 mM	7	Ateş et al. (2013)
Laccase	175.4 μmol/min	230.7 μmol/min	0.14 mM	0.13 mM	10	Birhanlı et al. (2022)
Laccase	175.4 μmol/min	97.4 μmol/min	0.14 mM	0.03 mM	10	Birhanlı et al. (2022)
Lipase	232 Umg^{-1}	238 Umg^{-1}	28 $mmolL^{-1}$	30 $mmolL^{-1}$	5	Facin et al. (2018)
β-Glucosidase	43 Umg^{-1}	61 Umg^{-1}	5.4 $mmolL^{-1}$	4.3 $mmolL^{-1}$	3	Califano et al. (2018)
α-Amylase	34.48 Umg^{-1}	25 Umg^{-1}	0.45 $mmolL^{-1}$	0.53 $mmolL^{-1}$	10	Bindu et al. (2020)
Xylanase	2.23 Umg^{-1}	3.83 Umg^{-1}	8.45 $mmolL^{-1}$	1.51 $mmolL^{-1}$	8	Wong et al. (2020)
Horseradish peroxidase	0.66 U	0.52 U	5.5 mM	8 mM	10	Salah et al. (2024)
l-Asparaginase	119.2 U	270.6 U	2.09 $mmolL^{-1}$	2.76 $mmolL^{-1}$	10	Noma et al. (2021)
Pectinase	31.5 U	43.67 U	0.81 $mmolmL^{-1}$	0.89 $mmolmL^{-1}$	7	Ladole et al. (2021)
Naringinase	50.2 U	67.11 U	1.33 $mmolmL^{-1}$	1.38 $mmolmL^{-1}$	7	Ladole et al. (2021)
Inulinase	0.83 U	0.76 U	2.74 $mmolmL^{-1}$	1.88 $mmolmL^{-1}$	12	Torabizadeh et al. (2018)
Carbonic Anhydrase	1.6011 mM min^{-1}	5.0525 mM min^{-1}	7.6283 mM	21.549 mM	-	Chang et al. (2021)

Note: f = free enzyme, i = immobilized enzyme.

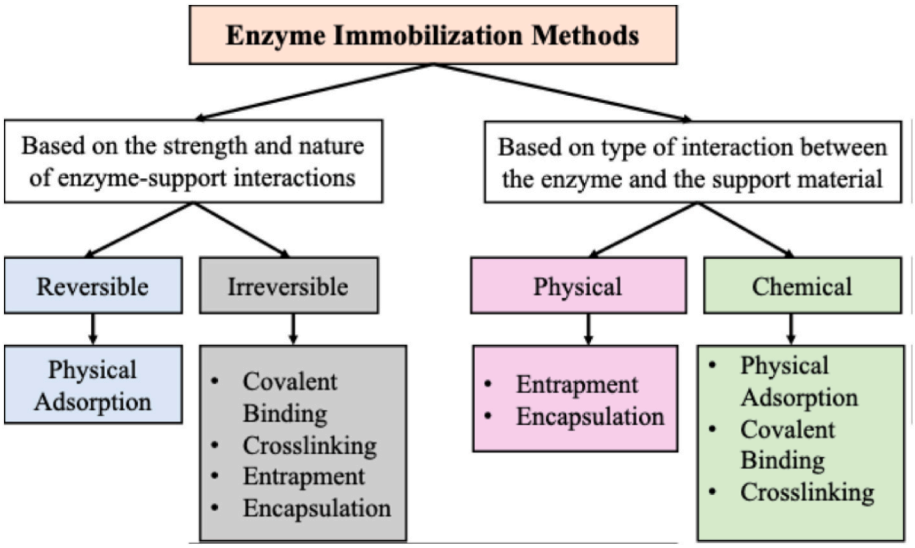


Fig. 3. Overview of methods of enzyme immobilization based on strength and nature of enzyme-support interactions as well as based on the type of interaction between the enzyme and the support material.

2016). Entrapment immobilization confines enzymes physically within a porous polymeric or fibrous network, using either covalent or non-covalent interactions. In this approach, enzymes are embedded within a three-dimensional lattice, typically generated through gelation, precipitation, or polymerization of natural polymers (such as alginate or chitosan) or synthetic materials (like polyacrylamide or silica). The pore size of the matrix can be carefully controlled to prevent enzyme leakage while still allowing substrates and products to diffuse freely. On the other hand, encapsulation, involves enclosing enzymes within semi-permeable membrane-bound vesicles. In this system, enzymes remain in solution but are spatially confined within micro-compartments. Unlike entrapment, encapsulation relies on selective semipermeable barriers that permit the passage of small molecules while retaining the larger enzyme molecules inside (Sirisha et al., 2016).

The success of immobilized enzymes depends on several interrelated parameters that reflect catalytic efficiency, stability, and cost-effectiveness under practical operating conditions. Since enzymes vary widely in their catalytic mechanisms, optimal conditions, and responses to immobilization, no single metric universally captures immobilized enzyme performance (Yuan et al., 2023). Nevertheless, certain analytical methods are consistently used across studies as summarized in Table 3.

Enzyme activity (U) is the most fundamental measure, representing the rate of substrate conversion or product formation, typically expressed in micromoles of substrate converted per minute under defined conditions. Specific enzyme activity (U mg^{-1}) normalizes this rate to the amount of enzyme protein, allowing comparison between different preparations or immobilization methods (Morshed et al., 2019). Immobilization yield (%) quantifies the proportion of total enzyme successfully attached or retained within the support matrix relative to the initial enzyme used, while retained activity (%) evaluates the catalytic activity of the immobilized enzyme compared to the free enzyme, reflecting the impact of immobilization on enzyme functionality. Enzyme loading (%) indicates the proportion of enzyme mass incorporated into the total immobilized catalyst, influencing both catalytic efficiency and cost (Morshed et al., 2021b). Catalyst consumption (g kg^{-1}), defined as the amount of enzyme required to produce a given mass of product, is a practical parameter for assessing process economics and scalability. Finally, space–time yield (STY; g L^{-1} h^{-1}), also known as productivity (kg g^{-1}), provides an overall measure of process performance by integrating reaction rate, enzyme stability, and operational duration (Yuan et al., 2023; Cornish-Bowden, 2013). In addition to these core parameters, extended metrics such as the half-life ($t_{1/2}$), deactivation rate constant (k_d), and turnover number (k_{cat}) are valuable for assessing enzyme longevity and intrinsic catalytic efficiency. Kinetic constants like the Michaelis constant (K_m) and the effectiveness factor (η) further help differentiate between diffusion-controlled and reaction-controlled processes. Parameters including the Thiele modulus (φ), reusability, and storage stability provide deeper insights into

Table 3
Common Analytical Method for Quantification of (Immobilized) Enzyme Performance (revised from Yuan, Y., Shen, J., & Salmon, S. (2023) (Yuan et al., 2023) under creative commons CC-BY license).

Metric	Description
Enzyme Activity (U) = $\frac{\mu mol\ of\ substrate\ converted\ (or\ product\ formed)}{Time\ (min)}$	Measures the catalytic rate of substrate conversion under defined conditions.
Specific Enzyme Activity ($U \cdot mg^{-1}$) = $\frac{U}{mg\ of\ enzyme\ protein}$	Normalizes enzyme activity per unit mass of enzyme protein.
Immobilization Yield (%) = $\frac{Amount\ of\ immobilized\ enzyme}{Total\ amount\ of\ enzyme\ used} x\ 100$	Indicates the efficiency of enzyme attachment to the support matrix.
Retained Activity (%) = $\frac{Immobilized\ enzyme\ activity}{Free\ enzyme\ activity} x\ 100$	Compares the catalytic performance of immobilized versus free enzyme.
Enzyme Loading (%) = $\frac{Mass\ of\ enzyme\ in\ support}{Total\ mass\ of\ immobilized\ material}$	Represents the proportion of enzyme incorporated into the immobilized catalyst.
Catalyst Consumption ($g \cdot kg^{-1}$) = $\frac{Amount\ of\ enzyme\ (g)}{Amount\ of\ product\ (kg)}$	Assesses process economy by relating enzyme usage to product yield.
Space–Time Yield ($g \cdot L^{-1} \cdot h^{-1}$) = $\frac{Amount\ of\ product\ (g)}{Reactor\ volume\ (L)\ x\ Unit\ time\ (h)}$	Reflects overall process productivity and operational efficiency over time.

enzyme robustness and practical reusability over multiple cycles. Collectively, these metrics form a comprehensive framework for comparing immobilization strategies and optimizing enzyme-based catalytic systems for industrial applications.

3. Progress and prospects of industrial application of immobilized enzymes

Enzymes are vital in many industrial applications due to their ability to catalyze specific chemical reactions efficiently. Through protein engineering, enzymes can be tailored for desired industrial purposes. They are widely employed in the detergent, textile, pharmaceutical, food industries, medical sensing as well as in wastewater treatment, CO_2 capture, and recycling as summarized in Table 4. In the food sector, immobilization enhances thermostability and resistance during processing, as demonstrated by alkaline protease on mesoporous zeolite/silica, which converted milk to cheese and retained 74% of its activity after 16 days. Similarly, amylase on chitosan exhibited improved thermal stability, pH resistance, and yield in barley hydrolysis, while immobilized pectinase clarified apple juice efficiently over multiple cycles (Kumari et al., 2015; Benucci et al., 2020; Raspopova et al., 2016; Cerreti et al., 2017). In detergents, enzyme immobilization improves stability and catalytic activity, aiding stain removal at ambient temperatures, with proteases, amylases, and lipases showing increased reusability. For example, protease bonded to Eudragit preserved wool tensile strength, and nanoenzyme conjugates retained activity after multiple cycles at 60 °C (Vasconcelos et al., 2006; An et al., 2014; Sharma et al., 2008). In textile processing, immobilized enzymes enhance stages such as desizing, bleaching, and biopolishing, improving both process efficiency and sustainability (Soares et al., 2011; Tzanov et al., 2002). In the pharmaceutical industry, immobilized enzymes are used for the production of active pharmaceutical ingredients, the synthesis of chiral compounds, and drug delivery systems, providing higher efficiency and stability in these processes (Basso et al., 2020). Environmental applications also benefit from immobilized enzymes, with laccase, tyrosinase, and manganese peroxidase effectively degrading pollutants like dyes and phenolic compounds, maintaining high activity across cycles. In parallel, enzyme immobilization underpins progress in biosensing and bioelectronics (Hadian et al., 2024; Ferraraccio et al., 2023). The coupling of enzymatic reactions with electrochemical or optical transduction enables rapid and selective signal generation. Moreover, enzyme-driven biofuel cells and bioelectronic devices offer sustainable energy options for wearable and environmental monitoring systems. Immobilized carbonic anhydrase is particularly effective in CO_2 capture, retaining over 80% activity after multiple uses, significantly boosting industrial CO_2 sequestration. Overall, enzyme immobilization enhances stability, reusability, and efficiency, making it a valuable tool across a wide range of industrial and environmental applications (Morshed et al., 2021b; Behary et al., 2025).

The future of immobilized enzyme technology is moving beyond traditional industrial use toward smart, multifunctional systems for environmental and health monitoring applications. Emerging research is increasingly focused on integrating enzymes with advanced materials such as nanofibers, conductive textiles, metal–organic frameworks, and hydrogels to create stable, reusable, and high-performance platforms (Khan, 2021). In environmental protection, immobilized enzymes are expected to contribute to next-generation pollutant removal systems by enabling continuous degradation of dyes, pharmaceuticals, microplastics, and greenhouse gases in water and air streams (Homaei et al., 2013). In parallel, enzyme immobilization is gaining importance in biosensing and bioelectronics. Enzymes such as glucose oxidase, laccase, and urease are being immobilized on wearable fabrics, flexible electrodes, and microfluidic devices to develop non-invasive biosensors for monitoring health biomarkers, toxins, and environmental contaminants (Morshed et al., 2021a). When combined with electrochemical or optical transduction, these systems enable real-time signal generation. Additionally, enzyme-based biofuel cells and bioelectronic devices are emerging as sustainable power sources for wearable electronics (Maghraby et al., 2023).

4. Enzyme immobilization on textiles to develop biocatalytic textiles

Immobilization of enzyme on textile was first introduced by Shemer et al., in 1979 (Morshed et al., 2021a). Since then, significant progress has been made in the domain in terms of use of various engineered textile as support materials for enzyme immobilization (Kiehl et al., 2015; Kahoush et al., 2022; Shim et al., 2017). Enzyme immobilization on textiles relies on creating stable interactions between enzyme molecules and the textile matrix, primarily through surface modification and chemical or physical attachment methods. The textile matrix serves as both a support and a microenvironment that influences enzyme orientation, conformational flexibility, and diffusion of substrates and products. The mechanism typically begins with the functionalization of textile fibers such as polyester, cotton, or acrylic using techniques like plasma activation, chemical grafting, or the incorporation of functional groups (e.g., amine, carboxyl) to introduce reactive sites for enzyme binding (Cristóvão et al., 2011; Mohamed et al., 2016; Almulaiky et al., 2023).

In our previous studies, we have reported the surface modification of polyester textile using polyethylenimine, chitosan, or dendrimers to achieve better immobilization yield, activity retention, and enzyme loading of enzyymes (Morshed et al., 2019, 2021a, 2021b). Our report (2021) highighed that, chitosan and polyethylenimine introduced amine-rich functional groups on the textile surface. These amine-rich functional groups resulted in high catalytic benefit of glucose oxidase enzyme and longevity with repeated use. Later on, Shen, J., Yuan, Y., & Salmon, S. (2022) has also reported the effectiveness of chitosan on immobilization of carbonic anhydrase enzyme on cotton (Shen et al., 2022a). Subsequently, a patent by Salmon, S., & Yuan, Y. (2023) reported enzyme entrapment within chitosan matrices using mild immobilization techniques that preserve the activity of acid-sensitive and heat-sensitive enzymes (Salmon et al., 2023).

All common immobilization methods including physical adsorption, covalent binding, crosslinking, entrapment/encapsulation has been

Table 4
Overview of industrial applications of immobilized enzymes.

Enzyme	Immobilization Method	Immobilization Support	Application	Ref.
Protease	Entrapment	Mesoporous silica and zeolite	Milk coagulant for dairy products	Kumari et al. (2015)
Protease	Crosslinking	Chitosan beads	Production of gluten-eliminated beer	Benucci et al. (2020)
Protease	Covalent binding	Eudragit	Blood and egg yolk stain removal	Vasconcelos et al. (2006)
Protease	Physical adsorption/Covalent binding	Mesoporous silica nanospheres	-	Ibrahim et al. (2016)
Amylase	Covalent binidng	Chitosan	Barley malt hydrolysis	Raspopova et al. (2016)
Amylase	Crosslinking	Glutaraldehyde	Starch stain removal	Shukla et al. (2016)
Pectinase	Encapsulation	Polyvinyl alcohol	Apple juice clarification	Cerreti et al. (2017)
Pectinase	Physical adsorption/ Crosslinking	Polymeric matrix	Apple juice clarification	Rajdeo et al. (2016)
Lipase	encapsulation	A-lactalbumin nanotubes	Flavor enhancement for low-fat cheeses	Guan et al. (2021)
Lipase	Physical adsorption/ Crosslinking	Arylamine glass beads	Oil stain removal	An et al. (2014)
Lipase	Covalent binidng	Glutaraldehyde	Oil stain removal	Sharma et al. (2008)
Lipase	Physical adsorption	Hydroxyapatite	Production of methyl acetate aroma	Trbojević Ivić et al. (2016)
β-galactosidase	Covalent binding	Chitosan/silica	Lactose hydrolysis in dairy products	Ricardi et al. (2018)
β-galactosidase	Covalent binding	Polyvinyl alcohol	Lactose hydrolysis in dairy products	Batsalova et al. (1987)
β-galactosidase	Encapsulation	Calcium alginate	Hydrolysis of cheese whey lactose	Mörschbächer et al. (2016)
Laccase	Crosslinking	Cu(II) ion chelated chitosan	Wastewater treatment	Bayramoglu et al. (2012a)
Laccase	Covalent binding	Glycidyl methacrylate functionalized polyacrylamide alginate	Wastewater treatment	Maghraby et al. (2023)
Laccase	Entrapment	Alginate bead	Wastewater treatment	Sondhi et al. (2018)
Laccase	Physical adsorption	TiO_2–ZrO_2–SiO_2	Wastewater treatment	Antecka et al. (2018)
Manganese peroxide	Covalent binding	Fe_3O_4/chitosan	Wastewater treatment	Siddeeg et al. (2019)
Manganese peroxide	Crosslinking	Chitosan beads	Wastewater treatment	Bilal et al. (2016)
Horseradish peroxide	Physical adsorption/ Crosslinking	Polyaniline grafted polyacrylonitrile	Wastewater treatment	Bayramoglu et al. (2012b)
Horseradish peroxide	Physical adsorption/covalent binding	Polyamide 6	Wastewater treatment	Jankowska et al. (2021)
Tyrosinase	Crosslinking	Iron oxide nanoparticles	Wastewater treatment	Abdollahi et al. (2018)
Tyrosinase	Crosslinking	Polyacrylonitrile beads	Wastewater treatment	Wu et al. (2017)
Tyrosinase	Physical adsorption	Sodium aluminosilicate	Wastewater treatment	Seetharam et al. (2003)
Laccase	Covalent binding	Zeolite	Wastewater treatment	Ameri et al. (2021)
Laccase	Physical adsorption	Sodium zeolite Y	Wastewater treatment	Taghizadeh et al. (2020)
Laccase	Covalent binding	Cu(II)-chitosan	Wastewater treatment	Alver et al. (2017)
Laccase	Covalent binding	Epoxy functionalized silica	Wastewater treatment	Mohammadi et al. (2018)
Laccase	Crosslinking	Magnetic nanoparticles	Wastewater treatment	Qiu et al. (2020)
Cellulase	Physical adsorption	Polyvinyl alcohol coated chitosan beads	Textile Processing	Dinçer et al. (2007)
Pectinase	Encapsulation	Sodium bis-2-ethylhexysulpho-succinate	Textile Processing	Sawada et al. (2001)
Glucose oxidase	Covalent binding	Alumina and glass	Textile Processing	Tzanov et al. (2002)
Horseradish peroxide	Covalent binding	Biocarbon	Wastewater treatment	Qiu et al. (2020)
Glucose Oxidase	Physical adsorption	Nowoven textile	Wastewater treatment	Morshed et al. (2019)
Glucose Oxidase	Physical adsorption	Nowoven textile	Wastewater treatment	(Morshed et al., 2021b; Behary et al., 2025)
Glucose Oxidase	Physical adsorption	Carboxymethyl chitosan/polyaniline/multi-walled carbon nanotubes	Medical sensing	Hadian et al. (2024)
Lactate dehydrogenase	Encapsulation	Alginate hydrogels	Medical sensing	Ferraraccio et al. (2023)
Carbonic anhydrase	Physical adsorption	Cotton textile	CO_2 capture	Shen et al. (2022a)
Carbonic anhydrase	Covalent binding	Geopolymer microspheres	CO_2 capture	Chang et al. (2021)
Carbonic anhydrase	Covalent binding	Polyethyleneimine/dopamine co-deposited MOFs	CO_2 capture	Xv et al. (2022)

explored for immobilizing enzymes on textile (Kiehl et al., 2015, 2017; Gao et al., 2014; Xu et al., 2013, 2014; Almulaiky et al., 2019; Mohamed et al., 2013; Li et al., 2011). Key factors influencing enzyme immobilization on textiles include the surface charge, hydrophilicity, porosity, and roughness of the textile matrix, which collectively govern enzyme orientation, binding affinity, and accessibility of the active site. The nature of the enzyme–support interaction ranging from weak physical adsorption to strong covalent bonding plays a decisive role in determining enzyme stability, resistance to leaching, and long-term reusability (Morshed et al., 2021a). The physicochemical microenvironment around the immobilized enzyme, particularly local pH, ionic strength, and temperature, can induce conformational changes that alter catalytic efficiency (Khan, 2021; Molina-Fernández et al., 2021; Sathishkumar et al., 2014). Moreover, mass transfer phenomena such as substrate diffusion and product release within the immobilized layer critically influence observed kinetic parameters and overall reaction rates.

Immobilized enzymes on textiles offer significant application

potential by combining biocatalytic efficiency with the versatility of textile materials. Such fabrics have been explored for environmental remediation, including degradation of dyes, pollutants, and CO_2 capture through carbonic anhydrase-functionalized textiles (Sathishkumar et al., 2014; Temoçin et al., 2018; Parmar et al., 2024; Xiao et al., 2023). In biomedical applications, enzyme-functionalized textiles exhibit antimicrobial, anti-biofouling, and wound-healing properties (Morshed et al., 2019; Parmar et al., 2024; Xu et al., 2021). They also enable self-cleaning, deodorizing, and catalytic degradation of organic contaminants under mild conditions. Integration with responsive or conductive fibers has further expanded their use in biosensing and smart wearable technologies (Kahoush et al., 2022).

Taken together from the above description of available works offers critical insights into the interplay between enzyme chemistry, support architecture, and operational performance. Textiles emerge not only as mechanically flexible and scalable supports but also as tunable microenvironments capable of modulating enzyme orientation, conformational dynamics, and local reaction conditions. Surface functionalization strategies, particularly those introducing amine-rich or polysaccharide-based interfaces have proven effective in enhancing enzyme loading and catalytic longevity; however, their influence on enzyme structure–function relationships remains insufficiently understood. While strong enzyme–support interactions improve resistance to leaching and operational stability, excessive rigidity or steric hindrance can compromise catalytic turnover, underscoring the need for rational immobilization design rather than empirical optimization alone. Moreover, mass transfer limitations, heterogeneous binding sites, and microenvironmental gradients within fibrous matrices continue to obscure accurate kinetic interpretation and scale-up predictions. Despite promising demonstrations in environmental remediation, biomedical textiles, and emerging areas such as CO_2 capture, most studies remain proof-of-concept, with limited evaluation under continuous operation or realistic field conditions. Standardized methodologies for comparing immobilization efficiency, enzyme lifetime, and regeneration remain notably absent. Addressing these gaps will require systematic, interdisciplinary approaches that integrate textile engineering, enzyme engineering, and reaction engineering. Such efforts are essential for establishing robust structure–property–performance relationships and for advancing enzyme-functionalized textiles from laboratory innovations to deployable, sustainable technologies.

5. Biocatalytic textiles for water purification application

Water pollutants such as dyes, pharmaceuticals, as well as Phenolic endocrine-disrupting chemicals (PEDCs) from domestic, agricultural, or industrial activities, represent some of the most pressing environmental challenges due to their persistence and toxicity. These substances pose significant risks to both human health and ecosystems and are often difficult to remove or degrade using conventional methods. Consequently, biocatalytic textiles has emerged as a promising approach for wastewater treatment. Examples of various biocatalytic textiles in pollutant removal are summarized in Table 5.

5.1. Dyes removal

Studies have demonstrated that biocatalytic textile can effectively degrade a variety of dyes and pigments, offering advantages such as enhanced stability and reusability. For instance, Kolak et al. (2023) (Kolak et al., 2023) reported immobilizing laccase from on PCL/PEI electrospun fibers, enhancing enzyme stability and reusability. The immobilized fibers efficiently decolorized Orange II and Malachite Green (up to 86% and 46%) and degraded 2,6-dichlorophenol (up to 64.5%) under optimized conditions. Arica et al. (2017) reported the successful immobilization of laccase onto a fibrous polymer-grafted polyvinyl chloride support and assessed its catalytic efficiency in the degradation of three representative dyes: Procion Green H4G, Brilliant Blue G, and Crystal Violet. The immobilized enzyme system demonstrated superior decolorization performance compared to the free

Table 5
A summary of removal of pollutant from wastewater using biocatalytic textiles.

Pollutant name	Pollutant category	Enzyme used	Pollutant removal, %	Reusability	Ref.
Chlortetracycline, Carbamazepine, and Diclofenac	Pharmaceuticals	*Laccase*	48.6-72.7%	10	Taheran et al. (2017a)
Reactive Black 5, Reactive Blue 114, Reactive Yellow 15, Reactive Yellow 176, Reactive Red 239, and Reactive Red 180.	Dyes	*Laccase*	~90-96% (individual) -45-63% (mixed)	2	Cristóvão et al. (2011)
Remazol blue RR	Dyes	*Glucose oxidase*	93.0%	6	Kahoush (2019)
Methyl parathion, Coumaphos	Pesticides	*Organophosphate hydrolase*	~100%	-	Gao et al. (2014)
Staphylococcus epidermidis, *Escherichia coli*	Pathogenic contaminants	*Glucose oxidase*	99.9%	-	Morshed et al. (2019)
Paracetamol	Pharmaceuticals	*Horseradish peroxidase*	83.5%	7	Xu et al. (2015)
Bisphenol A	PEDCs	*Laccase*	90.0%	-	Dai et al. (2016a)
2,4,6-trichlorophenol	PEDCs	*Laccase*	85.0%	10	Xu et al. (2013)
Triclosan	PEDCs	*Laccase*	65.0%	-	Xu et al. (2014)
Phenol	PEDCs	*Horseradish peroxidase*	29.68%	25	Temoçin et al. (2018)
Chlortetracycline	Pharmaceuticals	*Laccase*	58.3%	7	Taheran et al. (2017b)
Pentachlorophenol	PEDCs	*Horseradish peroxidase*	83.0%	8	Niu et al. (2013)
Bisphenol A, Triclosan and 2,4-dichlorophenol	PEDCs	*Laccase*	92.6 - 99.7%	10	Dai et al. (2016b)
Methyl orange	Dyes	*Horseradish peroxidase*	98.0%	10	Arslan (2011)
Naproxen, diclofenac	Pharmaceuticals	*Laccase*	90%	5	Zdarta et al. (2019)
2,4,6-trichlorophenol	PEDCs	*Laccase*	95.4%	7	Xu et al. (2017)
Catechol	PEDCs	*Laccase*	39.0%	-	Wang et al. (2014)
Bisphenol A	PEDCs	*Tyrosinase*	80.0%	10	Zdarta et al. (2020)

laccase, confirming the advantages of immobilization in enhancing dye removal efficiency (Morsy et al., 2020). In another report by Katarzyna et al. 2021 (Jankowska et al., 2021), a sustainable dye removal method has also been introduced using horseradish peroxidase immobilized on electrospun polyamide 6 fibers via adsorption and covalent bonding. The optimized biocatalytic materials effectively decolorized Reactive Black 5 and Malachite Green in simulated polluted seawater, achieving over 70% removal within 60 min (Jankowska et al., 2021). Sathishkumar, Palanivel et al. (2014) reported that laccase was immobilized on cellulose nanofibers to decolorize reactive dyes and simulated dye effluent, achieving enhanced thermal and pH stability, optimal performance with HBT as a mediator, and over 85% decoloration efficiency sustained for up to five reuse cycles. In another study, glucose oxidase was immobilized on plasma-treated carbon felt using the natural crosslinker genipin, forming a stable bio-catalyst that retained 40% activity, achieved up to 93% dye discoloration and 34% COD removal in Bio-Fenton and Bio-electro-Fenton processes, demonstrating its potential for sustainable dye removal from water (Kahoush et al., 2021). Our previous studies have also demondstrated the effective remoal of dyed using glucose oxidase enzyme immobilied on polyester textiles (Morshed et al., 2019, 2021b).

5.2. Pharmaceuticals and pesticides removal

Pharmaceutical and pesticides compounds are continuously released into the environment and have been increasingly detected in wastewater systems. Due to their complex chemical structures and biological activity, they pose substantial risks to aquatic ecosystems and human health. Therefore, developing efficient methods for the removal of pharmaceutical and pesticide residues from wastewater is of great importance. Among various approaches, enzymatic degradation particularly using immobilized enzymes on textile supports has emerged as a promising and sustainable solution. Several studies have demonstrated the potential of immobilized enzyme–textile systems for pharmaceutical and pesticides removal (Temoçin et al., 2018). For instance, Taheran, Naghdi et al. (2017) (Taheran et al., 2017a) immobilized laccase onto electrospun nanofibrous textiles through covalent bonding and achieved significant degradation of widely detected pharmaceuticals such as chlortetracycline, carbamazepine, and diclofenac. Within 8 h of batch treatment, degradation efficiencies reached 72.7%, 63.3%, and 48.6% for chlortetracycline, carbamazepine, and diclofenac, respectively. Moreover, the immobilized biocatalyst could be reused for up to 10 cycles with stable performance. In a subsequent experiment, the same researchers applied immobilized laccase in a continuous flow system, achieving 58.5% removal of chlortetracycline while retaining over 50% of the enzyme's initial activity after seven reuse cycles. They also noted that higher flow rates reduced removal efficiency, with optimal results observed at a flux of 3 mL/h•cm^2. Similarly, Gao et al. (2014) (Gao et al., 2014) reported that textiles functionalized with immobilized organophosphate hydrolase effectively degraded common pesticides in wastewater. These enzyme-functionalized textiles were integrated into a column reactor and were able to operate continuously for over 60 days at high flow rates without significant loss of catalytic efficiency. In another study, Xu et al. (2015) (Xu et al., 2015) developed a multifunctional textile by immobilizing horseradish peroxidase onto electrospun poly(vinyl alcohol)/poly(acrylic acid)/SiO_2 nanofibers, enabling both adsorption and catalytic degradation of paracetamol. Their system achieved an 83.5% removal rate and maintained 40.6% of the enzyme's initial activity after seven cycles. Kinetic analysis indicated that paracetamol degradation followed first-order reaction kinetics. Furthermore, Zdarta et al. (2019) (Zdarta et al., 2019) utilized electrospun poly(L-lactic acid)-co-poly(ε-caprolactone) nanofibers to immobilize laccase for the treatment of wastewater containing anti-inflammatory drugs like naproxen and diclofenac. Under optimized conditions, their system achieved up to 90% removal efficiency for both pollutants at concentrations up to 1 mg/L. Horseradish peroxidase was also immobilized in sodium alginate capsules and poly (vinyl chloride) electrospun fibers, achieving 100% immobilization and over 80% activity retention with minimal changes in kinetics. The immobilized enzyme efficiently degraded pharmaceuticals, removing over 80% of sulfamethoxazole and carbamazepine at 1 mg/L and over 40% at 10 mg/L under mild conditions. It also demonstrated high reusability and storage stability, retaining over 60% activity after 20 days and 10 cycles, unlike the free enzyme (Zdarta et al., 2021).

5.3. Phenolic endocrine-disrupting chemicals (PEDCs) removal

Among the emerging strategies for wastewater treatment, enzyme-immobilized textile materials have shown significant promise for the elimination of Phenolic endocrine-disrupting chemicals. These systems combine the catalytic action of enzymes with the structural and adsorption capabilities of textile substrates, delivering a synergistic effect in pollutant removal. Several studies have demonstrated the efficiency of immobilized laccase on textile-based supports for degrading phenolic contaminants through various immobilization route (see illustration in Fig. 4) (Chen et al., 2022). Xu, Chi et al. (2013) (Xu et al., 2013) reported that laccase immobilized on electrospun fibrous textiles was capable of removing over 85% of 2,4,6-trichlorophenol from wastewater under optimized conditions. This removal was attributed to both enzymatic degradation/mineralization and adsorption onto the fibers. Notably, the immobilized enzymes retained 50% of its activity after ten operational cycles, indicating reasonable reusability. In a study, Xu et al. (2017) (Xu et al., 2017) also immobilized laccase onto a polyacrylonitrile/polyvinylidene fluoride nanofibrous textile. This system achieved 95.4% removal of 2,4,6-trichlorophenol within 270 min and maintained 65.9% of its initial activity over seven reuse cycles. Similarly, Xu et al. (2014) (Xu et al., 2014) used mesoporous nanofiber textiles as supports for laccase to remove triclosan, while Wang, Cui et al. (2014) (Wang et al., 2014) demonstrated catechol removal using laccase immobilized on polyacrylonitrile/montmorillonite/graphene oxide composite fibers. In these cases, pollutant removal resulted from a combination of adsorption onto the fibrous matrix and enzymatic transformation (Xu et al., 2013, 2017; Wang et al., 2014; Morshed et al., 2020).

Further advancements were made by Dai, Yao et al. (2016) (Dai et al., 2016a), who co-encapsulated laccase and multi-walled carbon nanotubes in electrospun nanofibrous membranes using an emulsion electrospinning technique. These membranes effectively removed up to 90% of bisphenol An under a wide range of environmental conditions, such as pH, temperature, and initial concentration. Another study by the same group developed carbon nanotube-modified laccase-coated fibrous membranes that could degrade various phenolic pollutants, including bisphenol A (92.6 ± 0.74%), triclosan (95.5 ± 0.46%), and 2,4-dichlorophenol (99.7 ± 0.02%). Impressively, the membranes sustained high removal efficiency over ten consecutive cycles without significant deactivation. Besides laccase, other enzymes have also been explored for immobilization onto textile substrates. Temoçin, Inal et al. (2018) (Temoçin et al., 2018) immobilized horseradish peroxidase (HRP) covalently onto electrospun poly(vinyl alcohol)–polyacrylamide nanofibrous textiles for phenol removal from real wastewater. The system achieved a 29.68% removal rate within 180 min and could be reused for up to 25 cycles while maintaining 54% of its original activity. Likewise, Niu et al. (2013) (Niu et al., 2013) encapsulated HRP within electrospun nanofibers to remove pentachlorophenol, achieving 83% removal at ambient temperature. More recently, Zdarta et al. (2020) (Zdarta et al., 2020) optimized the immobilization of tyrosinase using response surface methodology. The immobilized enzyme was able to remove approximately 80% of bisphenol A within 90 min at pH 7 and 25 °C, demonstrating excellent storage stability (90% activity retained after 30 days) and functional reusability (over 80% removal after 10 cycles). These studies demonstrate significant progress in enzyme immobilization on nanofibrous and textile substrates, resulting in highly effective,

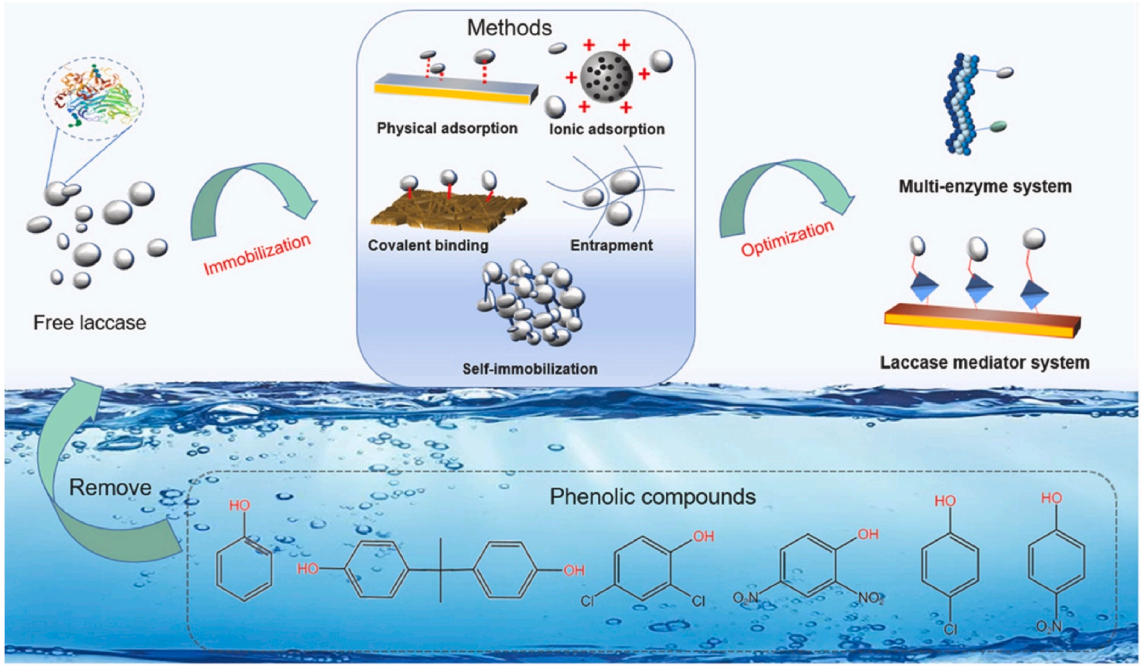


Fig. 4. Schematic illustration of utilization of immobilized laccase for the degradation of phenolic compounds in aqueous solutions (reused from Chen, Z., Oh, W. D., & Yap, P. S. (2022) (Chen et al., 2022) under creative commons CC-BY license).

reusable, and stable systems for removing various phenolic pollutants from water. The integration of enzymes like laccase, horseradish peroxidase, and tyrosinase with advanced materials such as carbon nanotubes and electrospun nanofibers offers promising, sustainable solutions for environmental remediation under diverse conditions.

5.4. Bacteria removal

Biocatalytic textile materials has proven to be an effective strategy for imparting strong antibacterial properties to fabrics, enabling their use in water disinfection and bacterial removal applications. Recent research demonstrates that immobilized enzymes such as cutinase, lysozyme, and pectinolytic enzymes, when anchored onto cotton or polyester-based textiles, can significantly inhibit the growth of both Gram-positive and Gram-negative bacteria, including common waterborne pathogens like *E. coli, Salmonella*, and *Staphylococcus aureus* (Parmar et al., 2024; Abdel-Gawad et al., 2024; Coradi et al., 2018). For example, cotton fabrics treated with immobilized cutinase and subsequently finished with antiseptic solutions exhibited clear inhibition zones against *S. aureus, MRSA, E. coli,* and *P. aeruginosa*, confirming their antibacterial efficacy. Similarly, immobilization of lysozyme and hydramacin-1 on cotton fibers resulted in up to 98% removal of *E. coli* and *Bacillus subtilis* from water, with the antimicrobial activity retained even after repeated washing and physical stress. In one of our study study, glucose oxidase was robustly immobilized on plasma- and chemically modified polyester nonwoven fabric, achieving up to 31% enzyme loading and 81% active immobilized enzyme with enhanced thermal stability and reusability over six cycles. The highlight is the successful integration of amine-functionalized surfaces enabling stable enzyme attachment, improved antibacterial activity against *Staphylococcus epidermidis* and *Escherichia coli*, and strong potential for bio-industrial and antimicrobial applications (Morshed et al., 2019). These findings highlight the potential of enzyme-immobilized textiles as reusable, eco-friendly, and highly effective materials for bacterial removal in water treatment and filtration systems (Xu et al., 2021; Ibrahim et al., 2007).

Collectively, the studies summarized in this section underscore the considerable potential of biocatalytic textiles as multifunctional platforms for wastewater treatment; however, they also reveal important limitations that currently constrain broader implementation. While high removal efficiencies have been demonstrated for dyes, pharmaceuticals, PEDCs, and bacterial contaminants, most reported systems rely on controlled laboratory conditions using synthetic wastewater, low pollutant concentrations, and short operational times. Consequently, the long-term robustness of these materials under realistic water matrices characterized by fluctuating pH, competing substrates, natural organic matter, and fouling agents remains insufficiently explored. In many cases, pollutant removal arises from a coupled mechanism of adsorption and enzymatic transformation, yet the relative contribution of each process is rarely quantified, complicating mechanistic interpretation and performance prediction. Reusability is often reported as cycle number rather than cumulative treatment capacity or enzyme turnover, limiting meaningful comparison across systems. Furthermore, kinetic analyses frequently overlook mass transfer constraints intrinsic to fibrous and nanofibrous architectures, resulting in apparent reaction rates that may not reflect intrinsic enzyme activity. Although integration into continuous-flow and column reactors represents a critical step toward practical deployment, such studies remain limited. Advancing biocatalytic textiles toward real-world wastewater treatment will require standardized benchmarking, validation under complex effluents, and explicit consideration of scalability, fouling resistance, and lifecycle sustainability alongside catalytic performance.

6. Biocatalytic carbon dioxide capture

Post-combustion, pre-combustion, and oxyfuel combustion are the three primary CO_2 capture process linked to various combustion processes as illustrated in Fig. 5. Carbon dioxide is extracted from the flue gas following fuel combustion in post-combustion CO_2 capture (Wall, 2007). During precombustion, the fuel; typically coal or natural gas is treated before it is burned (Olabi et al., 2022; Olajire, 2010). In oxyfuel combustion, fuel is burned using pure oxygen instead of air, which minimizes nitrogen in the exhaust and reduces thermal NO_x emissions (Buhre et al., 2005). The resulting flue gas mainly consists of CO_2, water vapor, particulates, and SO_2. Standard technologies can remove particulates and SO_2, leaving a gas stream with a high CO_2 concentration

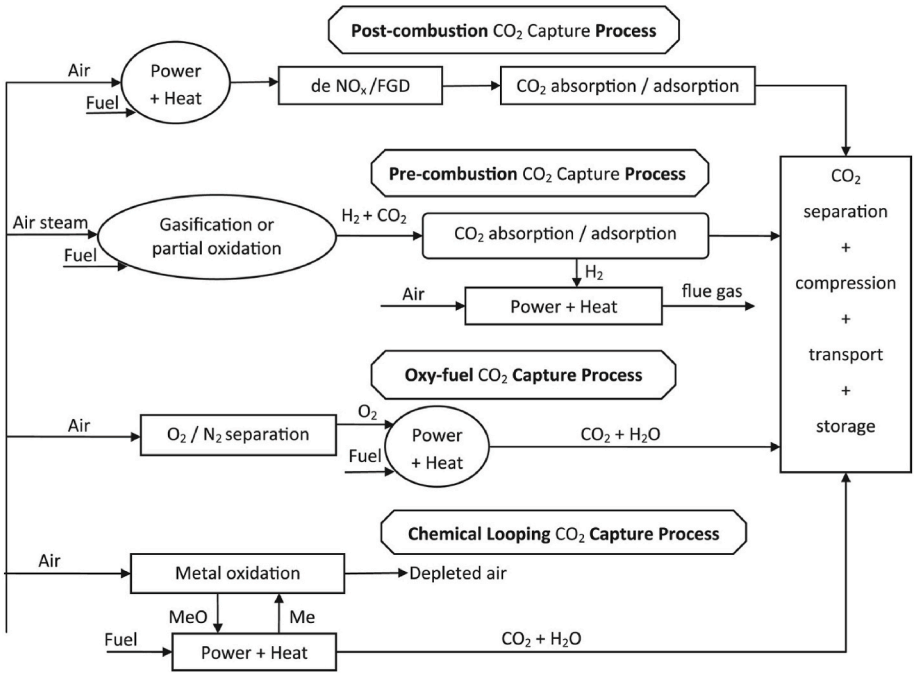


Fig. 5. Overview of main CO_2 capture systems associated with different combustion processes, namely, post-combustion, pre-combustion and oxyfuel combustion (used from Leung, D. Y., Caramanna, G., & Maroto-Valer, M. M. (2014) (Leung et al., 2014) under Creative Commons CC-BY license).

(typically 80–98%), suitable for compression, transport, and storage (Leung et al., 2014; Li et al., 2022). While the process is technically viable, it requires large amounts of oxygen from an energy-intensive air separation unit, leading to higher costs and an energy penalty of over 7% compared to conventional plants (Leung et al., 2014). Recently, integrating enzymes with CO_2 capture technologies has gained attention to enhance separation efficiency and facilitate CO_2 utilization, aligning with both carbon capture and storage (CSS) and the carbon capture and utilization (CCU) concept. Enzymes provide advantages over synthetic catalysts due to their operation under mild conditions and high specificity, enabling selective CO_2 conversion into value-added products with high purity. This biocatalytic approach is being explored in both absorption column systems and membrane-based technologies.

Different enzymes can speed up the process of turning CO_2 into something else, but only a few of these enzymes can be used in a separation unit to capture CO_2. The most widely applied enzymatic reaction in carbon capture is the conversion of CO_2 to bicarbonate, catalyzed by carbonic anhydrase (CA) (Hartanto et al., 2024; Ren et al., 2019). In biological systems, CA plays a crucial role in maintaining acid–base balance by rapidly interconverting CO_2 and bicarbonate. It is a lyase enzyme, meaning it breaks bonds without hydrolysis or oxidation, and is among the fastest enzymes known, with a turnover rate reaching up to 10^6 s^{-1}. It is a metallo-enzyme with an active site containing a zinc ion coordinated by three histidine residues and a water/hydroxide molecule (see Fig. 6) (Shi et al., 2015). Beyond CA, several other enzyme classes have been investigated for their potential to convert CO_2 into valuable chemicals such as decarboxylases, formate dehydrogenase (FDH), and engineered nitrogenase etc (Hartanto et al., 2024). Decarboxylases can catalyze the reverse decarboxylation process, incorporating CO_2 into organic substrates (Leung et al., 2014; Glueck et al., 2010). These enzymes provide an efficient route for CO_2 fixation under mild conditions, avoiding the harsh pressures or temperatures often required in chemical catalysis. For instance, phenylpyruvate decarboxylase and acetyl-CoA carboxylase have demonstrated the ability to capture CO_2 and convert it into key intermediates for biochemical synthesis. Such enzymatic carboxylation reactions are increasingly being explored for sustainable carbon capture, recycling and biomanufacturing applications. In addition to that, the reduction of CO_2 to formate or formic acid using formate

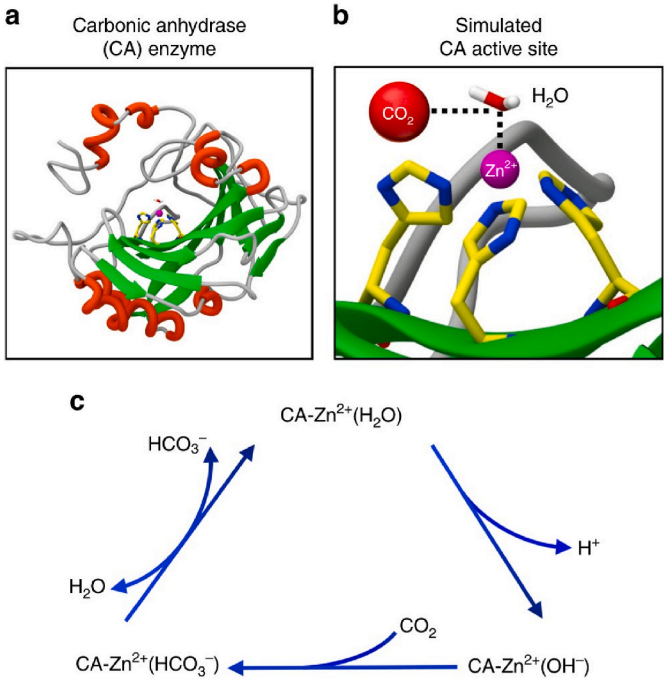


Fig. 6. Carbonic anhydrase enzyme and its CO_2 capture mechanism. (a) Ribbon structure of CA. (b) Active site with a zinc ion coordinated by three histidines and a water molecule. (c) Catalytic cycle converting CO_2 to HCO_3^-, driven by concentration gradients. The rate-limiting step is deprotonation of the zinc-bound water (reused from Fu et al. (2018) (Fu et al., 2018) under Creative Commons CC-BY license).

dehydrogenase, an oxidoreductase is has also been reported as a promising system (Calzadiaz-Ramirez et al., 2022). This reaction effectively transforms CO_2 into a stable, energy-rich liquid fuel or chemical feedstock that can serve as a hydrogen carrier. Metal-containing FDHs, typically found in anaerobic microorganisms, possess sophisticated

active sites with molybdenum (Mo) or tungsten (W) atoms coordinated to molybdopterin guanine dinucleotide cofactors, along with amino acid residues such as histidine and arginine that stabilize intermediates and facilitate electron transfer (Jayathilake et al., 2019; Amao, 2018). These cofactors enable multi-electron redox reactions, allowing efficient reduction of CO_2 even at ambient conditions. The electron donors often include reduced ferredoxin or nicotinamide cofactors (NADH/NADPH), linking the reaction to biological energy cycles. Similarly, engineered nitrogenase enzymes, which naturally reduce nitrogen to ammonia, have been modified to reduce CO_2 into methane (CH_4) under ambient conditions, using ATP and electron donors as reducing equivalents. These biocatalytic processes offer environmentally friendly alternatives to industrial chemical reduction methods, operating under mild pH and temperature while maintaining high specificity and turnover rates. Owing to their remarkable catalytic efficiency, these CO_2-converting enzymes hold great potential for integration into large-scale carbon management strategies.

Depending on the target application, these biocatalysts can be employed as single-enzyme systems for direct CO_2 conversion or assembled into cascade-enzyme systems that perform sequential reactions to yield higher-value carbon products. (Amao, 2018; Liao et al., 2022). One example of cascade system involves converting CO_2 into methanol catalyzed by multiple dehydrogenase enzymes (FDH, formaldehyde dehydrogenase ($F_{ald}DH$), and alcohol dehydrogenase (ADH)) (Marpani et al., 2017). Another cascade converts CO_2 into oxaloacetate through initial hydration of CO_2 to bicarbonate mediated by CA, followed by carboxylation of phosphoenolpyruvate using phosphoenolpyruvate carboxylase (Park et al., 2016). This pathway is attractive due to its cofactor-free operation and the versatility of bicarbonate as an intermediate for downstream enzymatic or mineralization reactions. Overall, while cascade enzymatic systems demonstrate high selectivity and sustainability potential for carbon capture and utilization, challenges remain in cofactor dependency, reaction efficiency, and process scalability (Leung et al., 2014). Current enzymatic CO_2 capture technologies with free enzymes often suffer from thermal deactivation in high-temperature desorbers, necessitating frequent enzyme replenishment. To tackle this, immobilization of enzymes, particularly carbonic anhydrase, has emerged as a promising strategy for efficient and sustainable carbon dioxide capture. CA catalyzes the rapid hydration of CO_2 to bicarbonate, a reaction central to many biomimetic carbon capture processes. However, the free form of CA is limited by instability, high cost, and lack of reusability, making immobilization essential for industrial applications (Molina-Fernández et al., 2021). Supports like geopolymer microspheres, metal-organic frameworks (MOFs), diatom biosilica, poly(ionic liquid) membranes and textile structure have been explored, each offering unique advantages in terms of enzyme loading, activity retention, and operational stability (Wen et al., 2020; Shen et al., 2022a; Molina-Fernández et al., 2021; Ren et al., 2019; Wang et al., 2024). The following section discusses the carbon capture by using immobilized CA enzyme on textiles (biocatalytic textiles).

6.1. *Carbon dioxide capture by enzyme immobilized biocatalytic textiles*

Biocatalytic textiles is slowly gainiing popularities for sustainable carbon capture, combining the high catalytic efficiency of biocatalysts with the robustness, low pressure drop, flexibility, and processability of textile materials. Several studies have reported the immobilization of CA on various textiles support materials (Yuan et al., 2023; Shen et al., 2022a, 2022b; Xiao et al., 2023; Salmon et al., 2025; Sahoo et al., 2015; Abdul Wasay et al., 2025). Techniques such as covalent bonding, physical adsorption, layer-by-layer assembly, and entrapment using polymeric coatings have been employed to anchor enzymes onto textiles while preserving their catalytic sites. Shen, J. et al. (2022) (Shen et al., 2022b) highlighted that, durable textile structured packings with covalently immobilized carbonic anhydrase were developed as efficient, "drop-in-ready" catalysts for CO_2 absorption in environmentally benign solvents (see Fig. 7). The system exhibited excellent CO_2 capture efficiency, retaining full performance for 71 days and 85% after one year of storage, while maintaining stability during 500 h of continuous operation—demonstrating robust, versatile, and long-lasting biocatalytic materials suitable for industrial CO_2 capture and direct air capture applications. Immobilized CA onto textile provides high surface area, excellent mass transfer, and structural flexibility, which allows the enzyme to remain confined in the cooler absorber region where it contributes maximally to CO_2 hydration. A multiphase computational fluid dynamics model developed to investigate CO_2 absorption in a biocatalytic gas/liquid contactor using enzyme-immobilized textiles shows that, enzyme effectiveness was found to strongly depend on film thickness and hydrodynamics, providing critical insights and predictive metrics for optimizing falling film absorption and biocatalytic reactor design (Abdul Wasay et al., 2025). Xiao, Min et al. (2023) (Xiao et al., 2023) reported, carbonic anhydrase was immobilized on textile structured packing modules, achieving ~1.5 × higher CO_2 absorption and ~80% enzyme activity retention without replenishment. This approach improves CO_2 capture efficiency while potentially reducing absorber size and operational costs, demonstrating the benefits of immobilized CA for durable solvent-based CO_2 capture. Another study shows, carbonic anhydrase was immobilized on textile-based contactors within a rotating packed bed reactor to enhance CO_2 absorption from ambient air using aqueous K_2CO_3 solutions. Results shows immobilized enzyme significantly improved direct air capture efficiency, demonstrating the potential of biocatalytic textile contactors for process intensification in CO_2 capture systems (Salmon et al., 2025). Thermophilic carbonic anhydrase from *Sulfurihydrogenibium yellowstonense* was also covalently immobilized on electrospun PAN–PET–PAN nanofibers, yielding highly stable and reusable biocatalytic materials for CO_2 sequestration. The immobilized enzyme showed remarkable thermal stability (up to 89.8% activity retention at 60 °C), enhanced $CaCO_3$ formation (up to 5.8-fold over free enzyme), and strong tolerance to NOx and SOx, highlighting its potential for durable, pollution-resistant carbon capture applications (Effendi et al., 2020). Thereafter, bovine carbonic anhydrase was immobilized on electrospun polymer nanofibers using an enzyme precipitate coating (EPC) method, achieving exceptionally high stability with 65.3% activity retained after 868 days and strong resistance to metal chelation. The highly loaded EPC effectively converted CO_2 to bicarbonate, which accelerated microalgae growth by up to 231%, demonstrating its potential for durable, high-performance enzyme-mediated CO_2 conversion and utilization applications (Jun et al., 2020). Sahoo, Prakash C. et al. (2015) (Sahoo et al., 2015) reported, electrospun poly(lactic acid) membranes were modified with graphene oxide (GO) and nanosized metal-organic frameworks (nMOFs) to immobilize carbonic anhydrase, achieving substantially higher enzyme loading (~1.83 × and ~1.57 ×) compared to unmodified membranes. The modifications also enhanced the thermal stability, storage stability, and

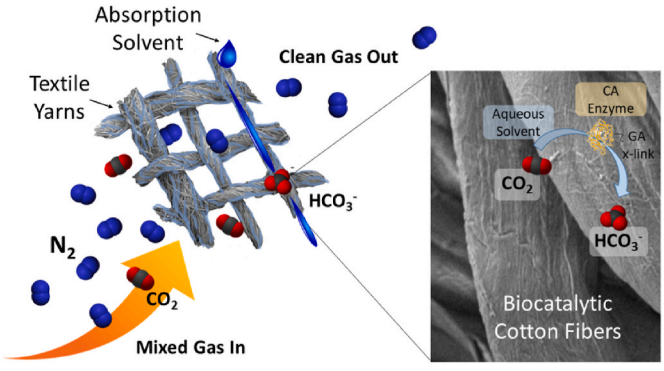


Fig. 7. Schematic illustartion of carbon dioxide capture by biocatalytic cotton textiles (reused from Shen, J. et al. (2022) (Shen et al., 2022b) under Creative Commons CC-BY license).

reusability of the immobilized CA, highlighting their potential for durable industrial CO_2 conversion applications.

Beyond CA, multi-enzyme systems immobilized on textiles are being explored to convert captured CO_2 into value-added products such as formate, methanol, or bicarbonates, linking capture with utilization. A study by Ji et al. (2015) (Ji et al., 2015) showed that, cationic polyelectrolyte-doped hollow nanofibers were fabricated via coaxial electrospinning to coencapsulate carbonic anhydrase, four other enzymes, and cofactors for efficient methanol synthesis from CO_2. The system achieved a record-high methanol yield of 103.2%, retained ~80% activity after 10 reuse cycles, and demonstrated that polyelectrolyte scaffolds can effectively stabilize cofactors and enhance multienzyme cascade efficiency for sustainable CO_2 conversion.

6.2. *Influencing factors of carbon capture by enzyme immobilized biocatalytic textiles*

Biocatalytic textiles provides a scalable strategy to CO_2 capture, with performance influenced by multiple factors of enzyme immobilization such as enzyme positioning, immobilization method, textile architecture, enzyme loading, and operational conditions (Sahoo et al., 2015). (a) The effectiveness of enzyme for CO_2 capture on textile surface is highly dependent on the positioning and accessibility of the enzyme (Morshed et al., 2021a). Optimal catalytic performance is achieved when CA is immobilized at or near the surface of textile fibers, where the enzyme can interact directly with both the gaseous CO_2 and the aqueous solvent. Enzymes buried deeper within the fiber matrix may experience limited mass transfer, as CO_2 must diffuse through the material to reach the active site, which significantly reduces overall reaction rates. Since CA is a diffusion-limited biocatalyst, this surface localization is essential to overcoming kinetic constraints and ensuring high absorption efficiency (Sahoo et al., 2015). (b) In addition to that, the immobilization strategy and material properties of the textile further influence enzyme performance. Covalent attachment to fiber surfaces generally provides superior mechanical stability and prolonged activity during repeated cycles of use, storage, and solvent exposure, compared to simple entrapment within polymer matrices, which can shield the enzyme but also restrict substrate access (Shen et al., 2022a). (c) Textile architecture plays a critical role: hydrophilic fibers promote efficient liquid transport and facilitate gas–liquid contact, while smaller fiber diameters, higher porosity, and rougher surfaces increase the relative surface area available for enzyme immobilization, enhancing accessibility of CA active sites. These factors collectively improve mass transfer and CO_2 capture rates. (d) Operational stability and durability are additional key considerations. Immobilized CA on textiles has demonstrated remarkable resilience, retaining over 80% of initial activity after hundreds of hours of continuous operation, multiple washing cycles, or long-term storage, highlighting its suitability for industrial CO_2 capture systems. Furthermore, textile-based biocatalysts maintain high activity across a range of CO_2 concentrations and benign solvent systems, offering versatility for different capture scenarios (Yuan et al., 2023). (e) Enzyme loading must be carefully optimized. While increasing the amount of CA can initially enhance absorption, excessive loading may result in overcrowding of the fiber surface and partial burial of active sites, reducing overall effectiveness. Strategic placement of the enzyme at the fiber surface is therefore more critical than maximizing total enzyme content. By balancing enzyme positioning, immobilization method, fiber architecture, and loading, textile-supported CA systems can achieve high catalytic efficiency, robust stability, and scalability, providing a promising low-energy and reusable approach for industrial CO_2 capture (Morshed, 2021; Morshed et al., 2021a; Maghraby et al., 2023).

6.3. *Post-combustion rector design for carbon capture by biocatalytic textiles*

Post-combustion CO_2 capture typically relies on gas absorption using aqueous solutions of chemically reactive absorbents, such as amines, which interact with CO_2 through thermally reversible reactions. While this approach is well-established, its efficiency is limited by the kinetics of CO_2 hydration and mass transfer between the gas and liquid phases (Shi et al., 2015). To overcome these limitations, textile-structured packings or nanofiber membranes functionalized with immobilized enzymes such as carbonic anhydrase have been introduced to enhance reactor performance. A typical process flow for CO_2 recovery using immobilized enyme, is illustrated in Fig. 8. Typically, in an absorber reactor, cooled flue gas containing CO_2 is passed through a column packed with either conventional structured packing or enzyme-functionalized textiles wetted by the absorbent solution. The immobilized enzymes act as catalytic sites at the gas-liquid interface, accelerating the conversion of CO_2 to bicarbonate (HCO_3^-), which is readily absorbed into the liquid phase (Xiao et al., 2023).

The high surface area, low pressure drop and porosity of textile packings improve liquid distribution and interfacial contact, maximizing the effective reaction volume and enhancing CO_2 uptake rates. Immobilization of enzymes on textiles not only accelerates absorption but also provides enhanced stability and reusability, addressing challenges associated with enzyme deactivation in homogeneous solutions. Enzyme-functionalized reactors maintain higher activity over longer operational periods, reducing the need for frequent replenishment and lowering operational costs (Yuan et al., 2023; Salmon et al., 2025). Mechanistically, the system leverages enzyme-mediated catalysis combined with optimized hydrodynamics, where liquid films flowing over the textiles allow CO_2 diffusion to active sites, and reaction rates are no longer limited by slow gas-liquid transfer. By integrating biocatalytic textiles into absorbers, CO_2 capture efficiency can be increased, residence time reduced, and absorber height potentially minimized, offering a scalable, energy-efficient strategy for post-combustion CO_2 capture (Technology, 2012).

7. Outlook and future perspectives

Biocatalytic textiles incorporating immobilized enzymes have emerged as a highly versatile and sustainable platform for environmental remediation, addressing critical challenges in both water purification and CO_2 capture. By combining the high catalytic selectivity and mild operational requirements of enzymes with the structural flexibility, high surface area, and tunable porosity of textile substrates, these materials offer several distinct advantages over conventional catalytic or

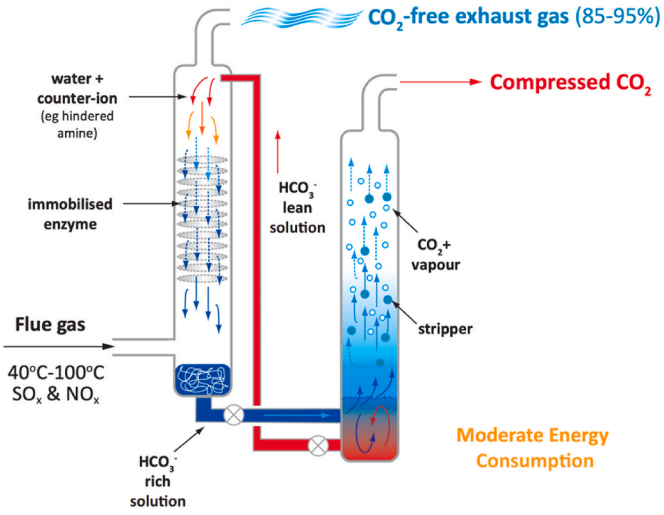


Fig. 8. Concept of a post-combustion rector for carbon capture by immobilized enzymes (reused from report on assessing post-combustion capture for coal-fired power stations in asia-pacific partnership countries (Technology, 2012) under Creative Commons CC-BY license).

sorption systems. The inherent modularity of textile-based platforms enables facile deployment in both batch and continuous-flow systems, while their adaptability to diverse reactor geometries facilitates enhanced liquid–enzyme and gas–liquid–enzyme interactions. Moreover, textile-based supports allow for strategic enzyme orientation, microenvironmental stabilization, and surface functionalization, enhancing both activity retention and operational longevity. Despite these advantages, the transition from laboratory-scale feasibility studies to industrial application faces significant scientific, engineering, and practical challenges. From a technical and operational perspective, mass transfer limitations remain a critical constraint. In water purification applications, pollutants must diffuse through the porous fibrous network to access immobilized enzyme active sites. This diffusion limitation is exacerbated in highly contaminated or viscous wastewater streams and can lead to substantial discrepancies between intrinsic enzyme activity and observed reaction rates. Similarly, in CO_2 capture, the diffusion of CO_2 to surface-bound carbonic anhydrase frequently becomes rate-limiting, particularly under high gas flow rates or turbulent conditions. These mass transfer constraints are compounded by the harsh conditions encountered in real-world applications. Enzymes are prone to deactivation due to thermal stress, pH fluctuations, ionic strength variations, and exposure to chemical inhibitors or fouling agents. For CO_2 capture, the high temperatures required in desorption units are incompatible with enzyme stability, necessitating innovative reactor designs that spatially separate enzymatic absorption from thermal regeneration. In water treatment, repeated exposure to oxidizing agents, detergents, and other industrial contaminants can rapidly diminish enzyme performance. Furthermore, the optimization of enzyme loading and spatial distribution on textiles is nontrivial. Excessive enzyme densities may result in steric hindrance, substrate diffusion barriers, and limited accessibility of active sites, whereas low loadings reduce catalytic throughput and economic feasibility.

Material selection and immobilization strategies further define the operational success of biocatalytic textiles. Textile properties such as fiber diameter, porosity, surface roughness, hydrophilicity, and mechanical strength significantly influence enzyme accessibility, substrate transport, and long-term durability. Covalent immobilization generally improves enzyme retention and reusability by minimizing leaching but may impose conformational constraints that reduce catalytic efficiency. Alternatively, polymer-based entrapment or coating strategies can protect enzymes from harsh environmental conditions but may exacerbate diffusion limitations and limit process rates. Hybrid strategies that combine covalent anchoring with microenvironmental stabilization, such as partial entrapment or surface functionalization, are increasingly recognized as promising solutions to balance activity, stability, and mass transfer. Scaling these systems to continuous-flow operations introduces additional complexity, as uniform enzyme distribution, reproducible immobilization, and mechanical integrity must be maintained over large surface areas and extended operational durations.

Future research should prioritize systematic, application-driven, and mechanistically informed studies that go beyond proof-of-concept demonstrations. Durability testing under realistic wastewater compositions, fluctuating environmental conditions, and extended operational periods is essential to evaluate true catalytic lifetimes. Kinetic characterization should account for mass transfer limitations and the coupled contributions of adsorption and enzymatic transformation to pollutant removal, enabling accurate modeling and predictive design. Hierarchically structured textiles with optimized porosity, surface chemistry, and wettability can enhance substrate accessibility while protecting enzyme functionality, particularly under continuous-flow or high-throughput conditions. For CO_2 capture, mobile or buoyant textile-based biocatalysts may maximize gas–liquid–enzyme interaction and facilitate rapid recovery and reuse, potentially reducing energy inputs associated with gas compression and absorption. High-throughput screening and computational modeling can accelerate the rational design of enzyme–textile systems, allowing simultaneous optimization of material properties, immobilization chemistry, and reactor architecture. From an industrial perspective, integration into modular water treatment or flue gas capture systems requires rigorous techno-economic and life-cycle analyses. Metrics such as cumulative treatment capacity, enzyme turnover, operational stability, regeneration requirements, and energy consumption should guide material selection, immobilization strategies, and reactor configuration. Life-cycle considerations—including raw material sourcing, recyclability of textile supports, and enzyme production impacts—will be critical for assessing the true environmental sustainability of these technologies. Finally, standardization of experimental protocols, including substrate concentrations, reactor geometry, and operational conditions, will enhance reproducibility, facilitate cross-study comparisons, and accelerate the translation of laboratory successes into scalable, economically viable processes.

In conclusion, biocatalytic textiles hold significant promise as a sustainable, versatile, and high-performance platform for water purification and CO_2 capture. Achieving industrial relevance will require a coordinated effort to address fundamental scientific questions regarding enzyme–material interactions, mass transfer limitations, and deactivation mechanisms, alongside engineering solutions that enable durable, modular, and scalable deployment. By combining advanced textile engineering, hybrid immobilization strategies, computational modeling, and rigorous life-cycle assessment, the field is well-positioned to transform biocatalytic textiles from laboratory innovations into low-energy, high-impact solutions for global environmental challenges. The convergence of materials science, enzyme biotechnology, and process engineering will be pivotal in realizing the full potential of this platform and establishing its role in sustainable water and carbon management technologies.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT in order to improve the language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Mohammad Neaz Morshed: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sweta Iyer:** Writing – original draft, Data curation, Conceptualization. **Vincent Nierstrasz:** Writing – original draft, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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